C. Monitor

time limit per test: 0.5 second
memory limit per test: 64 megabytes

Reca company makes monitors, the most popular of their models is AB999 with the screen size $a \times b$ centimeters. Because of some production peculiarities a screen parameters are integer numbers. Recently the screen sides ratio $x:y$ became popular with users. That's why the company wants to reduce monitor AB999 size so that its screen sides ratio becomes $x:y$, at the same time they want its total area to be maximal of all possible variants. Your task is to find the screen parameters of the reduced size model, or find out that such a reduction can't be performed.

Input

The first line of the input contains 4 integers — a, b, x and y ($1 \le a, b, x, y \le 2 \cdot 10^9$).

Output

If the answer exists, output 2 positive integers — screen parameters of the reduced size model. Output 0 0 otherwise.

Examples

input Copy

```
800 600 4 3
```

output Copy

```
800 600
```

input Copy

```
1920 1200 16 9
```

output Copy

```
1920 1080
```

input Copy

```
1 1 1 2
```

output Copy

```
0 0
```

Codeforces Beta Round 16 (Div. 2 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
370377832	Apr/09/2026 10:43	Accepted

→ Problem tags

binary search number theory *1800

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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